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RESEARCH ARTICLE

Prefix tuning with prompt augmentation for efficient financial news summarization

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Abstract

In financial markets, the sentiment expressed in news articles plays a pivotal role in interpreting and forecasting market trends, which also holds true for the task of financial news summarization (FNS). Leveraging AI models to analyze social science data, this paper employs financial sentiment to improve FNS effectiveness by introducing a novel method that combines the sentiment polarity extracted from financial news with prompt augmentation techniques to ensure that the generated summaries are emotionally consistent with the source articles. Specifically, the detected sentiments are embedded into prompts and provide directive instructions to the model to generate summaries. Furthermore, to address the problem of limited large-scale datasets for FNS and ensure more tailored results, we employed prefix tuning as a fine-tuning strategy. Preliminary results indicate that our combined methodology outperforms approaches that use only prefix tuning. The experimental findings further validate the significance of sentiment analysis in FNS, which enhances the accuracy of capturing and reflecting market sentiment, thereby yielding valuable insights into financial markets. This method not only improves the accuracy and relevance of summaries but also ensures that their content is emotionally consistent with the source news, offering a new perspective on financial news summarization.

Keywords Financial news summarization · Financial sentiment analysis · Natural language processing · Prompt augmentation · Prefix tuning

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Introduction

Markets flourish with new information, and there is a constant flow of new data. As many types of information (e.g., rumors, wiretaps, and scandals) can affect financial markets, financial news articles are considered a more stable and trustworthy source of information [1]. However, with constantly growing information, it is difficult for financial professionals and the general public to obtain timely and salient information from massive amounts of financial news, making financial news summarization (FNS) essential. FNS refines the original article into concise versions while retaining key factual details, sentiments, and perspectives. Compared to news summarization in other domains, FNS faces more complex challenges such as not only conveying facts accurately and unambiguously but also capturing the dynamic sentiments of the financial markets and potential market impacts [2]. This makes FNS a challenging and important research topic in natural language processing.

The evolution of general text summarization technology has been significant, shifting from simple and straightforward unsupervised extraction methods to the current deep learning (DL) architectures. With the advent of the transformer structure [3] and the bidirectional encoder representations from transformers (BERT) model [4], significant progress has been achieved in generative summarization methods, such as the Seq2Seq structure with the attention mechanism [5], text-to-text transfer transformer (T5) [6], and pre-training with extracted gap-sentences for abstractive summarization (PEGASUS) [7]. Despite the success of these DL models, several significant challenges remain. For example, the training of complex models often requires significant computational resources and massive annotated datasets [8]. For our FNS task, obtaining large datasets with manual summaries was difficult, limiting the performance of these models. Thus, the adaptability of these models to domain-specific tasks (e.g., financial news summaries) is yet to be explored. Furthermore, fine-tuning these models to specific tasks can lead to catastrophic forgetting of previously learned information. In light of these challenges, a novel approach, termed “Prefix Tuning" [9], has emerged as a promising solution. In this study, we apply it to the FNS task, to guide the model in conditional generation and enhance its adaptability without the drawbacks of traditional fine-tuning techniques, thus providing a tailored solution to the above issues.

Financial news is distinct from general news in its professional, qualitative, and specialized terminology. Beyond focusing on core facts, financial news delves into market sentiments, industry jargon, and the subtle effects on market dynamics [10]. For instance, a surge in the stock of a company may not only be attributed to plain data but may also be influenced by a favorable assessment or advantageous policy. These complex sentiments and industry-specific terms make sentiment analysis indispensable for interpreting financial narratives [11]. Integrating sentiment analysis into FNS tasks not only strengthens the summarization process but also deepens the insights gained from financial news articles, thereby improving market trend forecasts and investor sentiment understanding.

This study proposes an approach that integrates financial sentiment polarity into prompts, thereby guiding models to generate the corresponding summaries.

Embedding the sentiment label directly into the prompts makes the model more responsive to the nuances of financial news. To ensure efficiency, this method employs prefix tuning, which enhances the adaptability and performance of the model without the extensive resource demands typically associated with deep learning models. The synergy between sentiment-enriched prompts and prefix tuning represents a targeted and efficient strategy for financial news summarization. To the best of our knowledge, this is the first study to incorporate financial sentiment polarity into summarization prompts. Figure 1 compares our method with the existing approach, mapping the novel integration of sentiment analysis with the summarization process from initial news input to final summary output.

The experimental results show that the proposed approach achieves significant performance improvements over both prefix and fine tuning (baselines). Unlike other studies, the introduction of sentiment polarity as an intrinsic prompt combined with the prefixed-tuning technique enables the generation of more tailored financial news summaries. The remainder of this paper is organized as follows. In Sect. 2, the fundamental techniques used to explain the proposed method are briefly reviewed. Section 3 introduces the specific techniques of prompting and prefix tuning, which are integral to our approach, providing the necessary background and rationale for including them in the proposed method. Section 4 details our novel approach and explains how prompting and prefix tuning were integrated into our model to enhance the FNS task. Section 5 presents the experiments and results. Finally, Sect. 6 concludes the paper.

Literature review

Financial news summarization

Among the various types of news, financial news is a key source of information that influences global financial markets and economic decision-making. FNS is designed

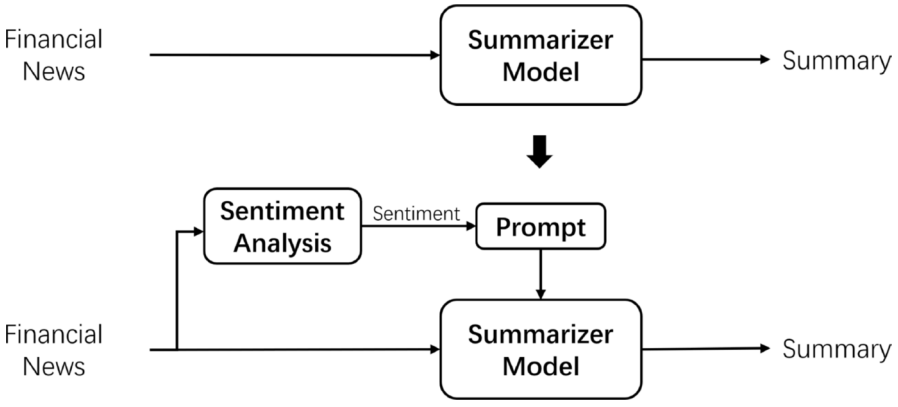


Fig. 1 Financial news summarization process

to refine the financial news into concise versions while retaining salient events, emotions, and opinions that affect the fluctuations in financial markets [12]. This offers valuable insights into understanding and forecasting financial markets, such as stock movements [13, 14]. Two mainstream approaches are used to address this task: extraction and abstraction.

Extractive methods produce summaries by selecting and ranking key sentences in a document based on their interpretation and relevance to the overall content. However, these methods do not generate new content, making them relatively simple and less demanding in terms of computational resources. This approach is notably applied to the financial narrative processing (FNP) task to summarize the annual reports of U.K. corporations [15, 16]. Graph-based unsupervised methods have been used to summarize stock news by ranking sentences within a graph based on their network structure [17, 18]. These methods assess the importance of sentences based on their connectivity and positional relevance in the graph, thereby preserving the original context and ensuring the inclusion of critical information in the summary. However, using graph-based techniques for summarizing financial news poses challenges because the salient content is always scattered across long-range interactions, which are difficult to represent effectively within a graph structure.

Abstractive summarization systems, on the other hand, build summaries by paraphrasing the text in the document. However, this method requires a deeper interpretation and understanding of the text. In [19], a hybrid extractive and abstractive summarization model was proposed to select relevant sentences using a graph-based approach, thereby utilizing a sequence-to-sequence abstractive model to generate a summary of financial news. In this manner, not only words can be generated from the vocabulary but also copy terms from the source text. However, such methods adopt a recurrent neural network (RNN)-based encoder, ignoring the long-range interaction information of financial news, thereby suffering from a long-term dependence problem [20].

Recently, strong efforts have been made to develop abstractive systems by focusing on encoder-decoder architectures pretrained on self-supervised methods [6, 7]. These can generate concise and coherent summaries that appear similar to those generated by humans. Yet these systems are prone to “hallucination", with the models generating content in the summaries that cannot be directly inferred from the source document. Moreover, the lack of large-scale, high-quality datasets is an urgent problem. Typically, in FNS tasks, reference summaries are prepared either manually or automatically in advance. However, the manual creation of such a vast number of reference summaries can incur high costs. This challenge is particularly pronounced in the finance domain, where crafting abstracts requires domain-specific expertise.

Sentiment in news summarization

In addition to hallucinations, factuality, and abstraction, studying sentiments is also important in summarization systems and in the corpora used to train them [21]. As summaries serve as an entry point for the source article, the sentiment they evoke

directly influences how users perceive the content, especially in financial news. However, there are few studies on summarization tasks that consider sentiments. We found that emotions in automatic news summarization were only studied in [21, 22]. In [21], the authors proposed an emotion-aware news summarization system utilizing a clustering module to group sentences based on topic and a sentence-ranking module to select the most relevant sentences for summarization, incorporating emotional words as query expansion terms to create emotionally aware summaries. The experimental results showed that emotion summaries contained more emotion words than the baseline system but did not show a significant improvement in the quality of the summaries. In [21], the prevalence and emphasis of emotions in news summarization corpora were also examined, evaluating how state-of-the-art abstractive summarizers capture and reflect these emotions in the generated summaries. Although these models can replicate the emotional densities, they often fail to match the emotional biases of human summarizers.

This study demonstrates that incorporating emotions into summarization tasks is crucial for creating summaries that are more consistent with human emotional responses; however, existing systems often struggle to accurately align these summaries with the emotional nuances observed in human-authored texts.

Proposed approach

In this paper, we propose a novel approach for an FNS task that does not require a large database. Our method focuses on developing models capable of controlling sentiment coherence more effectively, to help reduce hallucinations in the generated summaries. The proposed method makes two main contributions: (1) **Innovative direction in FNS** The sentiment polarity inherent in financial news is incorporated as a prompt for the first time. Our approach highlights the role of sentiment polarity in generating FNS, which ensures a more accurate representation of financial sentiment and market dynamics in the generated summaries. (2) **Efficiency and performance with prefix tuning on BART** Building on the bidirectional auto-regressive transformer (BART) [23] model, this study adopts the prefix-tuning methodology instead of the conventional fine-tuning method. Prefix tuning is a more parameter-efficient strategy, particularly in scenarios with limited data availability. This research leverages the large pretrained knowledge of BART by updating only the prefix part and adjusting it specifically for FNS. This approach not only ensures reduced computational costs but also prevents overfitting that might arise in limited-data scenarios.

Task-specific inputs: prompting & prefix-tuning

Prompting Prompting refers to pre-pending task-specific instructions and a few examples to the task input, thereby generating the output from the language model (LM), as originally demonstrated in [24]. In the field of natural language processing (NLP), prompting techniques play a pivotal role in model training and task-specific

adaptations, particularly when dealing with pretrained language models. First, there are 'hard' prompts [25], which are generated from interpretable words and tokens, and they must be hand-crafted by humans. The hard prompt remains unchanged during model training and inference phases, serving as a fixed instruction for model responses.

Unlike hard prompts, 'soft' prompts involve trainable embeddings or tokens that are appended to the model's input during training. These prompts are not directly interpretable as human-readable text but consist of a series of tunable parameters. Throughout the training process, these parameters are adjusted to optimize model performance on a specific task [26]. The above prompting methods only insert prompts into the bottom embedding layer of the pretrained LMs, whereas prefix tuning [9] adds soft prompts to all transformer layers to further increase prompting capacity.

Prefix tuning Alternatively known as continuous prompt tuning, prefix tuning was introduced as a method for conditional generation tasks to enable parameter-efficient transfer learning. This method appends tunable prefix vectors to the multi-head attention mechanism in every layer of a transformer, whereby two sets of prefix vectors, P_k and P_v, are concatenated to the original key K and value V matrices. Multihead attention is then performed on the new prefixed keys and values. The head is computed as in [27].

$$head = Attn(xW_q, concat(P_k, CW_k), concat(P_v, CW_v)) \tag{1}$$

where x is a query vector and C is a sequence of vectors over which we would like to perform attention; $K = CW_k$ and $V = CW_v$. This prefixed attention mechanism offers a more fine-grained control over the attention output, as evidenced by the altered computation of each attention head. This approach allows the attention mechanism to be dynamically modified, guiding the model to learn task-specific representations without overhauling the entire set of parameters. A similar idea, termed 'P-tuning' [28], introduces learnable prompt tokens to the input of pretrained LMs, emphasizing a minimalistic yet effective approach to task-specific adaptations without extensive parameter modification. However, compared to P-tuning that works only to a certain extent, the prefix-tuning approach offers a richer mechanism. As detailed in [27], the model achieves a balance between adaptability and complexity by altering the attention output in each layer using length-l prefix vectors, thus providing an edge in specific tasks.

In summary, methods such as prompting and prefix tuning bridge the gap between simplicity and efficiency.

Methodology

In this section, we present the proposed prefix-tuning prompt augmentation (PA) system for FNS tasks. This system is based on the prefix-tuning methodology proposed in [9]; with some modifications introduced into our prefix tuning-PA system for FNS tasks (Sect. 1).

In the financial domain, the sentiment polarity of news articles is pivotal to understanding financial news. Existing FNS methods may ignore this sentiment information, thereby generating summaries that are inconsistent with the original content sentiment. Our method integrates financial sentiment polarity into prompt and guides models to generate corresponding summaries that are aligned with the original sentiment of the financial news article.

Figure 2 provides a comprehensive overview of the proposed system. The components highlighted within the red dashed box in the figure represent the novel contributions of our proposed method, including the integration of sentiment polarity into the prompt and the modifications to the prefix-tuning process to better address the FNS task. Next, we present the key components and workflow of the prefix tuning-PA system, illustrating how it effectively integrates the FNS task.

FinBERT FinBERT [29] is a pretrained NLP model for analyzing the sentiment of financial texts. It was built by further training the BERT language model in the finance domain using a large financial corpus and fine-tuning it for financial sentiment classification. We chose the FinBERT model as our sentiment extraction module for financial news articles, primarily because of its exceptional performance in financial sentiment analysis. FinBERT demonstrated a high accuracy of 97% on a specific subset of Financial PhraseBank [30], a widely recognized dataset in the domain of financial sentiment analysis. This subset, characterized by 100% annotator agreement, represents the most reliable and clear example of sentiment in financial texts and provides a solid basis for evaluating the effectiveness of the model. By leveraging FinBERT's proven proficiency in accurately identifying sentiment polarity in financial contexts, we aimed to enhance the precision and relevance of the sentiment analysis of the proposed model for financial news processing. For a given financial news input x, sentiment polarity s is derived using FinBERT as follows:

$$s = FinBERT(x) \tag{2}$$

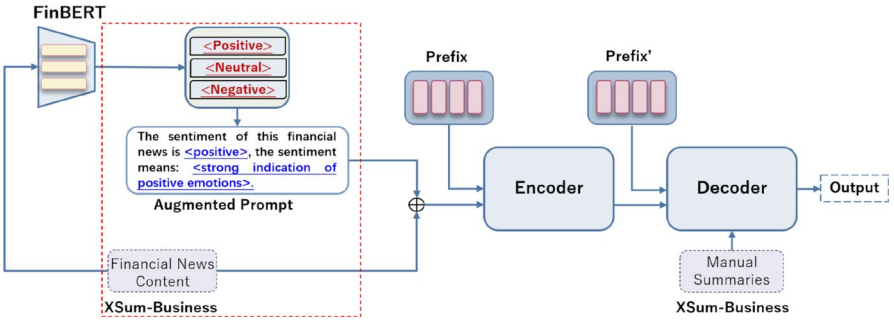


Fig. 2 The framework of our PREFIX-PA prompt augmentation model. Initially, FinBERT analyzes the sentiment polarity of financial news, which is embedded into the prompt sentence to generate summaries. Prefix tuning, involving the updateable vectors **Prefix** and **Prefix'**, is used to boost model efficiency, enabling the generation of summaries aligned with the determined sentiment. The components highlighted within the red dashed box represent the novel contributions of our proposed method, which extend existing approaches to better integrate sentiment polarity into the summarization process

Prompt augmentation Based on the extracted sentiment labels, an augmented prompt was designed for the FNS task, as detailed in [31]. This process involves adding to the fixed hard prompt an automatically selected sentiment label for each news item. The resulting 'Augmented Prompt' denoted as P_a, was effectively tailored to reflect the sentiment s of the news content. This concept is illustrated in Fig. 2, which shows the implementation of the augmented prompt in our system.

$$P_a = P_h(s) \tag{3}$$

This strategy varies the prompts based on sentiment labels to guide the output of the model in a more contextually relevant and nuanced manner. By leveraging the capabilities of the pretrained LM using the PA strategy, more accurate or specific outputs tailored to the sentiment of the FNS task were produced.

Encoder-decoder architecture We use the encoder-decoder architecture of BART to model $p_\phi(y \mid x)$, where x is encoded by the bidirectional encoder, and the decoder predicts y autoregressively, conditioned on the encoded x and its left context. BART is a pre-trained sequence-to-sequence model that combines a bidirectional Transformer encoder and an auto-regressive Transformer decoder, which excels at tasks like text summarization due to its robust denoising pre-training objective. In our experiments, we utilize the pre-trained BART model provided by Hugging Face as the base model. Importantly, the parameters of the BART model remain frozen throughout the training process to retain its pre-trained knowledge and reduce the risk of overfitting in limited-data scenarios. This setup allows our proposed prefix-tuning module to adapt the model to the financial news summarization (FNS) task with minimal parameter updates, leveraging the pre-trained model's extensive knowledge while ensuring computational efficiency.

Prefix parameter Inspired by prefix tuning [9], we prepended an FNS-specific prefix module to both the encoder and decoder to obtain $z = [PREFIX;x;PREFIX';y]$, where x is the article; y is the summary; and z represents the sequence of tokens. x is encoded by the bidirectional encoder, and y is predicted autoregressively by the decoder. The prefix parameters are stored in a trainable matrix P_θ, where i represents the position in the sequence, and the activation h can be formulated as

$$h_i = \begin{cases} P_\theta[i, :], & \text{if } i \in Prefix \\ LM_\theta(z_i, h_{<i}), & \text{otherwise} \end{cases} \tag{4}$$

Here, LM_θ represents the LM parameterized by θ. $LM_\theta(z_i, h_{<i})$ computes the activation of the current token z_i using its past context $h_{<i}$. P_θ is parameterized using a smaller matrix P'_θ and a feedforward multilayer perceptron (MLP) neural network because directly updating it can render it numerically unstable. After training, only the prefix P_θ is retained.

$$P_\theta[i, :] = MLP_\theta(P'_\theta[i, :]) \tag{5}$$

Model training using prefix tuning By applying the prefix-tuning method, the knowledge and capabilities of the large LM can be retained while specializing it for an FNS task. For the training objective, the main model parameters are fixed,

and only P_θ is trained. This helps achieve a balance between leveraging pretrained knowledge and refining the model to accommodate the specific summarization cues introduced in the sentiment polarity prompts.

The entire process can be described as the output of the summarization model $\hat{y}$, influenced by news input x, task-specific prefix P_θ, and augmented prompt P_a.

$$\hat{y} = Model(x, y, P_\theta, P_a) \tag{6}$$

The training loss is defined as the difference between the true summary y and model-generated summary $\hat{y}$.

$$\min_{P_\theta} \{Loss(y, \hat{y})\} \tag{7}$$

The loss function, denoted by *Loss*, measures the divergence between the true and generated summary. During the prefix-tuning phase, the main objective is to reduce the loss associated with the prefix P_θ, whereby the gap between the actual summaries and those generated by the proposed model is narrowed under the condition that only the prefix P_θ is optimized. Thus, summaries that not only accurately capture the core information of financial articles but also maintain the same sentiment context as the original on the FNS task, are generated.

Experiment

Datasets and metrics

For the FNS task, we used a subset of the XSum dataset [32] for business topics (0.06% of the original dataset). In the XSum dataset, each article is prefaced with an introductory sentence (i.e., summary) that is professionally written, typically by the author of the article. The topic of each article is filtered based on its uniform resource locator (URL) to identify business-related news. Based on the official splitting results, the final dataset was divided into training, validation, and test sets with 12,327 (90%), 730 (5%), and 710 (5%) articles, respectively.

To evaluate the content quality of the model-generated summaries, we utilized Recall-Oriented Understudy for Gisting Evaluation (ROUGE) [33], a widely used set of metrics and software package for assessing automatic summarization and machine translation systems in natural language processing. ROUGE compares generated summaries against human-produced reference summaries, focusing on n-gram and longest common subsequence overlaps. In this work, we report ROUGE-1, ROUGE-2, and ROUGE-L scores.

Furthermore, to provide a more comprehensive evaluation of the semantic quality of the generated summaries, we incorporated BERTScore [34] as a complementary metric. BERTScore computes similarity scores at the token level using contextual embeddings from pre-trained language models (e.g., BERT). Unlike ROUGE, which relies on exact n-gram matches, BERTScore captures semantic similarity, making it particularly effective for evaluating paraphrased or semantically similar outputs.

In our experiments, we report the F1 BERTScore, offering a holistic assessment of both lexical overlap and semantic fidelity.

Results and analysis

In the experimental setup, the XSum-Business dataset was used to conduct comparative analyses, focusing on two baseline methodologies, fine-tuning and prefix-tuning, both utilizing the BART model as a foundation. For the fine-tuning baseline, the pretrained fine-tuned BART model was employed for summarization tasks, as detailed in [23]. This model was further fine-tuned on our XSum-Business dataset for an additional 30 epochs. This additional training phase was to adapt the model more closely to the specifics of our dataset, potentially enhancing the summarization performance in the context of business news.

To evaluate the impact of the proposed approach, experiments were conducted under two conditions for each methodology: one employing our novel sentiment prompt and the other without it. This design allowed us to evaluate the effectiveness of incorporating sentiment polarity from financial news to guide the model to generate more reliable summaries, thereby ensuring a fair comparison by highlighting the differential impact of adding sentiment prompts in the context of both prefix and fine-tuning.

Comparison of summarization results Table 1 presents the results of the experiments, showcasing the performance of both baseline and sentiment-enhanced methods. The proposed augmented-prompt approach achieved consistent improvements across all metrics compared to the baseline methods, with the impact being more pronounced in the prefix-tuning setup.

- **Fine-tuning results** FINE-TUNE-PA showed modest improvements over the baseline, with increases of 0.52, 0.21, and 0.30 in R1, R2, and RL scores, respectively, and a slight gain in BERTScore-F1 from 90.16 to 90.35. The limited performance gain is due to the small size of the XSum-Business dataset (0.06% of the original), as fine-tuning relies heavily on large datasets to effectively adapt pretrained models, underscoring its data-intensive nature.

Table 1 Comparison of baseline with the proposed model. FINE-TUNE-PA and PREFIX-PA refer to the prefix- and fine-tuning methods with prompt augmentation. Both prefix- and fine-tuning underperformed our method

	R1	R2	RL	BERTScore-F1
FINE-TUNE	37.27	15.10	29.81	90.16
FINE-TUNE-PA (ours)	**37.79**	**15.31**	**30.11**	**90.35**
PREFIX	38.14	15.94	31.11	90.41
PREFIX-PA (ours)	**38.82**	**16.68**	**31.97**	**90.52**

Results in bold indicate outcomes from our proposed methods, with underlined scores highlighting the best performance across all evaluations

- **Prefix-tuning results** PREFIX-PA achieved notable improvements, with increases of 0.68, 0.74, and 0.86 in R1, R2, and RL scores, respectively, and a rise in BERTScore-F1 from 90.41 to 90.52. This demonstrates the effectiveness of incorporating sentiment prompts in enhancing summarization quality.

Overall, the PREFIX-PA outperformed all other approaches, validating the efficacy of our sentiment-based prompt augmentation strategy. The results confirm that the proposed method not only enhances the ROUGE scores but also achieves higher semantic similarity as reflected in the BERTScore-F1, with PREFIX-PA achieving the highest score (90.52).

Qualitative examples Table 2 lists the qualitative examples. In the first example, the augmented prompt plays a role in abstract generalization. The prefix baseline, which does not incorporate sentiment polarity, focuses on the specifics of export growth and attributes the surge to the strong demand from specific regions such as the US and Southeast Asia, whereas our method offers a macro perspective that correlates export growth with the broader economic recovery narrative in China. Moreover, the similarity between the proposed approach and the references is that both emphasize economic optimism, suggesting that augmented prompts can guide the model to generate summaries that are consistent with the sentiment tone of the original content. The second example is similar to the first; therefore, we do not provide a detailed explanation.

Table 2 The blue words in bold represent the salient information of the reference summary captured by the proposed method but ignored by the baseline. The first two lines represent good examples, and the third line is a bad example

Sentiment: Positive
Reference: Chinese exports rose more than expected in February, adding to optimism over **a recovery in its economy.**
PREFIX: China's exports jumped more than expected in February, helped by strong demand from the US and South East Asia.
Ours: China's exports jumped more than expected in February, indicating **a recovery in** the world's second largest **economy.**
Sentiment: Negative
Reference: Oil and gas exploration firm Tullow Oil has reported a pre-tax loss of £1.3bn for the year to 31 December as **low oil prices** bit into revenues.
PREFIX: North Sea oil firm Tullow Oil has reported a loss of £1.2bn for 2015.
Ours: Tullow Oil has reported a loss of £1.2bn for 2015 as it continues to struggle with **low oil prices.**
Sentiment: Positive
Reference: Sales and profits slipped at Boeing last year, **driven largely by lower deliveries of its military aircraft.**
PREFIX: Boeing shares have risen after the US aerospace giant reported a drop in sales and profits for 2015.
Ours: Boeing has reported a drop in fourth-quarter sales and profits **as demand for its commercial planes continued to slow.**

For the third example in the table, the generated summary leaned towards highlighting negative aspects, such as the decline in sales and profits caused by reduced military aircraft deliveries although the sentiment prompt provided to the model was positive. This discrepancy between the sentiment of the prompt and the content of the summary can be attributed to two reasons: (1) The target summaries may have emphasized similar downturns during training, which may have caused the model to be more attuned to these aspects. (2) The original text may be complex, including multiple positive and negative information points. In this case, the model may not have assigned adequate negative weights to phrases such as 'Sales and profits slipped,' resulting in a summary that contradicts the provided sentiment prompt. This observation underscores the influence of source content complexity on summary generation.

Ablation study In the ablation study, the impact of sentiment polarity on the effectiveness of our model was assessed. We removed the sentiment labels, as shown in blue in Fig. 2, and implemented a hard prompt: 'Prompt: Provide a concise and accurate summary of the financial news, including key events, factors influencing the market, and potential economic impacts.' This approach intentionally omitted sentiment data to discern the influence of direct and unambiguous prompt instructions on the output of the model.

As shown in Table 3, the ROUGE scores and BERTScore for the different hard prompt settings were as follows: for 'FINE-TUNE-HP', the scores were 37.02, 14.85, 29.28 and 90.20, while for 'PREFIX-HP', the scores were 38.12, 16.11, 31.23 and 90.45, respectively, for the prompt elements. In contrast, the 'PREFIX-SP' configuration, which solely utilized sentiment labels as prompts, yielded scores of 37.92, 15.78, 30.38 and 90.35, respectively. Using hard prompts and using only sentiment labels, both performed worse than the proposed methods ('FINE-TUNE-PA' and 'PREFIX-PA') which incorporate sentiment-augmented prompts. The decline in scores upon the removal of sentiment labels reinforces their significance in enhancing summary quality, underscoring the importance of considering sentiment in the FNS task.

Prefix length The effect of prefix length on the summarization task was explored using the prefix-tuning method, as shown in Fig. 3. The results revealed that the scores of both ROUGE-2 (16.68) and ROUGE-L (31.97) metrics achieved peak performance at a prefix length of 200, indicating that this is the ideal length for our task. However, there was a noticeable decline in performance when the prefix length

Table 3 Ablation study: 'FINE-TUNE-HP' and 'PREFIX-HP' configurations employ hard prompts, with the former applying it to fine-tuning and the latter to the prefix-tuning approach. Conversely, 'PREFIX-SP' utilizes the prefix-tuning framework with sole focus on sentiment information in the prompts, without any additional words

	R1	R2	RL	BERTScore-F1
FINE-TUNE-HP	37.02	14.85	29.28	90.20
PREFIX-HP	38.12	16.11	31.23	90.45
PREFIX-SP	37.92	15.78	30.38	90.35

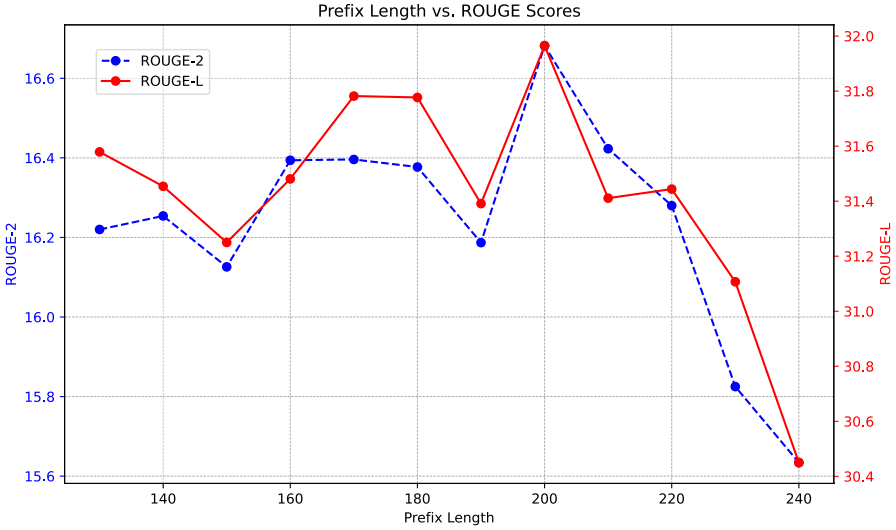


Fig. 3 Prefix Length vs. summarization performance. Performance increases as the prefix length increases up to a threshold of 200 and then a slight performance drop occurs. Two metrics (on two vertical axes) are reported in the plot

exceeded 200. This implies that excessively long prefixes can introduce unnecessary noise and detract the model from essential cues. Similarly, the performance suffers when the prefixes are less than 200 in length, highlighting the significance of an optimal prefix length for capturing pivotal task-specific cues. In conclusion, although prefix length in prefix tuning has a clear effect on model performance, it is not solely dependent on dataset size and seems to be closely tied to task categories, model design, and the nature of the data.

Conclusion

In this study, we presented an innovative approach for the FNS task, uniquely incorporating the sentiment polarity intrinsic to financial news as a prompt, thereby enriching the contextual foundation for a more precise summarization. To the best of our knowledge, this is the first initiative of its kind in this field. A prefix-tuning method based on the BART model was utilized to effectively harness BART's extensive pretrained knowledge, tailoring it to the specific subtleties and demands of financial summarization by selectively fine-tuning the model prefixes. Our preliminary findings substantiate the hypothesis that integrating sentiment analysis into FNS tasks not only refines the summarization process but also enriches the insights extracted from financial news articles. This enhances the predictive capabilities of market trends and investor sentiments.

The proposed methodology demonstrated impressive performance even in data-limited scenarios by leveraging the sentiment polarity of financial news as a directive prompt combined with prefix tuning. However, this approach has some limitations.

A significant constraint is the reliance on FinBERT as a sentiment extraction model, which, despite its advanced capabilities, cannot guarantee absolute accuracy in sentiment determination. This is particularly true of financial narratives, which are often complex and multifaceted. Thus, future research should consider exploring multi-aspect sentiment-analysis techniques to capture the nuanced sentiment dimensions prevalent in intricate financial narratives more comprehensively.

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Data availability The data used in this study for summarization are openly available in Hugging Face at https://huggingface.co/datasets/EdinburghNLP/xsum.

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no Conflict of interest.

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